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REPORT

Differing lesion recovery rates of two Caribbean stony coral species across a shallow water to mesophotic depth gradient suggest different sensitivity to repeated disturbance

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Abstract Following major stress events such as storms, bleaching events, or disease outbreaks, surviving corals must regenerate tissue to recover. We aimed to understand how this recovery changes across depth, hypothesizing that deeper corals would regenerate more slowly and that this may limit resilience to acute stressors. Two species of reef-building coral, *Orbicella franksi* (an intermediate-depth species) and *Agaricia lamarcki* (a depth-generalist), were tagged at selected sites across their overlapping depth range of 13–41 m and directly monitored for recovery from experimentally generated lesions across time. Overall, recovery rates were distinct between species and across depths, with *O. franksi* recovery rates showing high variability and declining at depth. In contrast, *A. lamarcki* maintained similar rates of recovery across the examined depth range. The consistent response of *A. lamarcki* suggests that it can attenuate its biology with changing light resources to maintain healing abilities in different environments. Recovery rates were additionally compared against environmental and biological covariates and it was found that only increased initial lesion size had a significant positive effect on tissue regeneration rates for *A. lamarcki*. Collectively, this suggests that some mesophotic coral reefs, despite having high coral cover, may be slower to recover from stress events if dominated by non-depth-generalist species, such as *O. franksi*, resulting in increased vulnerability to repeated stress events.

Keywords Mesophotic · Recovery · Tissue · Lesions

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Introduction

Study of coral recovery is crucial in the continued understanding and conservation of coral reefs, some of the world's most biodiverse and productive ecosystems (Hoegh-Guldberg et al. 2007; Moberg and Folke 1999; Plaisance et al. 2011). As the foundational species of coral reefs, scleractinian coral recovery from stress events is essential to the function of the reef ecosystem. In many cases, acute stress events, such as hurricanes, disease outbreaks, runoff, and bleaching events can affect individual coral colonies asymmetrically, and may cause death of tissue in only part of a coral colony (Fabricius 2005; Lugo-Fernández and Gravois 2010). This is known as "partial mortality", where the afflicted area, identified here as a "lesion", can sometimes be regrown from the remaining living tissue on the colony (Loya 1976). Tissue regrowth can be energetically expensive, and often requires re-allocation of energy away from other processes, such as calcification and reproduction (Kramarsky-Winter and Loya 2000; Meesters et al. 1994). Therefore, the rate of tissue regrowth can play a key role in overall reef resilience and recovery from major stress events.

Researchers have worked to identify key extrinsic and intrinsic factors that may affect tissue recovery. It is understood that lesion size, position, and shape inherently alter lesion recovery rates (Meesters et al. 1994, 1996). Recovery has also been shown to be species-specific (Hall 1997).

Because of the energetic demand associated with lesion recovery, coral health can also play a key role (Fine et al. 2002; Fisher et al. 2007; Ruiz-Diaz et al. 2016). Relating coral tissue recovery to the physical environment directly is less frequent (Nagelkerken et al. 1999; Sabine et al. 2015), and in these cases, recovery can be delayed or even halted by stressful events during or immediately before recovery (Henry and Hart 2005; Meesters and Bak 1993; Meesters et al. 1994). Therefore, relatively stressful physical regimes in an environment, such as temperature fluctuations (Carilli et al. 2012) and wave energy (Dollar 1982), may play an important role in recovery.

Deep-water or mesophotic coral ecosystems (MCEs), are light-dependent stony coral habitats found at depths deeper than about 30 m and are less studied relative to shallow reefs (Hinderstein et al. 2010; Tamir et al. 2019). These systems have different environmental conditions than shallow reefs (Lesser et al. 2009), and are more vulnerable to certain stressors compared to shallow water systems, such as destructive fishing practices and sedimentation (Smith et al. 2019a) and thermal stress (Smith et al. 2016b). Although coral tissue regeneration and lesion recovery has been extensively studied for shallow water reefs, the effect of increased depth on tissue recovery has only recently been investigated (Counsell et al. 2019). Mesophotic coral stress response and recovery from stress has not been effectively studied (Smith et al. 2019a).

Many stony coral species have depth ranges that extend from shallow to mesophotic depths (Carpenter et al. 2008; Laverick et al. 2018; 2019), and calcifying corals have been shown to alter skeletal structure to optimize light capture to the environment (Terán et al. 2010). In the Caribbean over 50% of shallow water coral species occur deeper than 30 m (Laverick et al. 2018). Two important reef-building coral species- the sheeting scroll coral *Agaricia lamarcki* and massive boulder star coral *Orbicella franksi*- can be found in both shallow and mesophotic depths, though each has distinct depth distributions across this range. *O. franksi* can be found as shallow as 3 m depth and as deep as 45 m, and often is the dominant reef-building species in the upper mesophotic (30–45 m) (Smith et al. 2010; Egan et al. 2021). By contrast *A. lamarcki* persists across a much wider depth range than *O. franksi*, occurring as shallow as 10 m and as deep as 80 m, notably to depths where many other scleractinian coral species are light-limited and cannot persist (Laverick, et al. 2017). The ability of *A. lamarcki* to act as a depth-generalist has largely been attributed to pairing with depth-specialized symbionts and increased use of heterotrophic feeding to maintain similar growth rates across its entire vertical range (Bongaerts et al. 2013; 2015; Crandall et al. 2016). As Agariciid species can maintain growth (Bongaerts et al. 2015), they are more likely to maintain recovery rates across their entire depth range. As a result, reefs dominated by *A. lamarcki* may be potentially more resilient to repeated stress events than reefs dominated by other species.

This study assessed coral tissue recovery for *O. franksi* and *A. lamarcki* across a shallow to mesophotic depth gradient to test the potential difference in resilience of a depth-specialist and depth-generalist coral species. A standardized lesion-recovery assay was conducted, assessing the ability of each species to recover and regenerate tissue across this depth range. Environmental regimes of temperature and benthic orbital velocity and coral biological characteristics were investigated as possible causes for observed differences in recovery across depth. By understanding differences and drivers in recovery for each species, a greater understanding of the future of MCEs can be reached in the face of continuing stress events.

Methods

Site selection

Corals were sampled at 7 different locations of varying depths from 13 to 41 m depth, previously identified by the Territorial Coral Reef Monitoring Program (TCRMP). To decrease the likelihood of coastal or other site-specific influences, all locations were located offshore from Saint Thomas, US Virgin Islands. In addition to a previously established TCRMP site at South Capella at 23 m depth, a new site in the nearby area was sampled separately as an additional (8th) site for this study. The new second South Capella site (referred to as "South Capella (deep)") was located in a deeper part of the reef in this same location (35 m). Collectively, these sites encompass the complete overlapping depth range of the two species, with the four deepest sites considered traditionally upper mesophotic (30–45 m depth), and the four shallowest sites considered shallow reefs (< 30 m depth). While *A. lamarcki* has been observed and even dominates in areas deeper than 41 m, *O. franksi* could not be sampled at these deeper depths due to low occurrence. All sites except the newly established deep site at South Capella have been monitored for coral health, diversity, and fish diversity annually since at least 2005. All data is publicly available at http://usvi-tcrmp.com/ (See Supplemental Figure S1 and Supplemental Table S1 for site and sampling day information).

Site monitoring

All sites were continuously monitored for seawater temperature using in situ subsurface temperature recorders (STRs) (HOBO Water Temp Pro v2, Onset Computer Corp., Bourne, MA). Probes have monitored temperature continuously at

15 min intervals at the research sites since 2010, with all data publicly reported annually online (accessed September 2018 at http://usvi-tcrmp.com/). These data were gathered and analyzed during the exact period of study at each location (For a full list of the precise study window of each location, see Supplemental Table S1). Temperature characteristics at each site were calculated as daily average, daily maximum, daily minimum, and daily standard deviation (summarized in Supplemental Tables S1–S3).

Water movement in the form of wave energy or benthic orbital velocity at each site was determined using a Simulated Waves Nearshore (SWAN) grid model reconfigured by the Caribbean Coastal Ocean Observation System (CariCOOS) for the US Virgin Islands, which calculated daily wave height, period, and wavelength for the closest SWAN grid area for each location during the sampling period (Booij et al. 1999). Two study locations, Buck Island and Flat Cay, were within the land-masked region in the original SWAN grid, and instead used a bilinear interpolation scheme to estimate the values at these locations based on the nearest corresponding SWAN grid points. Modeled and interpolated wave energy, subsequently, was translated to benthic orbital velocities (u) experienced by the corals at depth using linear wave theory (Smith et al. 2016a), given as

$$u = \frac{\frac{\frac{H}{2} * gT_{\mathrm{p}}}{L} * 1}{\cosh\left(\frac{2\pi d}{L}\right)} \tag{1}$$

where d is the local water depth at the study site, H is the significant wave height, Tp is the peak wave period, and L is the local wavelength corresponding to the peak wave period and the local water depth. Similar to temperature, the wave energy environment was fully characterized using daily maximum, minimum, average, and standard deviations of calculated benthic orbital velocities for each location during the study period. For minimum benthic orbital velocity, the smallest calculated nonzero value was used for analysis (see Supplemental Tables S1–S3).

Coral tissue lesion assessment

Lesion generation methodology replicated coral laceration regeneration assay methodology used in Sabine et al (2015), and developed in NOAA's Coral Disease & Health Consortium (NOAA 2016). Moving out in a random direction from a haphazardly placed marker at each site, colonies of *O. franksi* and *A. lamarcki* ($n = 3$ for each species at each site) greater than 10 cm in max diameter were identified and tagged using a cork tied to substrate using biodegradable twine. All colonies were upright, healthy, with no visible signs of bleaching or impaired health. After tagging, colonies were sized by length, width, and height (including any apparent old mortality), and a photo was taken of the colony surface facing downward using a Canon G1X camera under ambient light. Using a small chisel or flathead screwdriver to mimic physical damage, a single circle of tissue approximately 2.5 cm-diameter was removed from the center of the colony facing directly upwards. Particular care was taken to only remove tissue, with as little skeletal damage as practicable. While lesion generation methodology was intended to constrain lesions to a constant size, actual lesion areas ranged from 1.58cm^2 to 7.00cm^2. Precise lesion areas were later determined using scaled photos (explained in detail below) and factored directly into analyses (see "Statistical Analysis"). Removed tissue in lesion generation was retained for later symbiont genotypic analysis. A second photo was then taken of the inflicted area following lesion generation. Colonies were then left on the reef to recover and revisited approximately every 30 d for 120 d following initial lesion generation (see exact dates of sampling in Supplement Table S1). To assess differences in recovery rates through time recovery was calculated both between each sampling point and across the entire observation period. Colonies were re-photographed with a standardized scale bar until the end of sampling, the coral colony marker was lost, or the coral appeared completely healed.

Lesion photos were imported and analyzed in Fiji, a modified version of ImageJ2, a Java-based open-source image processing program. Using the polygon selection tool, a unique polygon was manually drawn around the apparent lesion during each sampling point. Because of the variable nature of the lesion generation, initial lesions were variable in size and shape. Lesion area and perimeter were calculated manually using the polygon selection tool and a scale bar placed in each photo. Lesion recovery was then measured three different ways to ensure recovery was accurately assessed. While linear extension of tissue is the standard method for measuring tissue movement in whole coral colonies, extension is highly variable in small, enclosing lesions. Instead, assessments of change in planar area capture any tissue recovery that occurred between observation periods (see Fig. 1).

The first metric assessed the decrease in size of the bare lesion area, considered the tissue regeneration rate (TRR). Lesions were manually re-traced in scaled photos at each timepoint using the polygon selection tool and measured for actual area in cm^2. The difference in area between these traces and the measured initial lesion area was considered the amount of regenerated or recovered tissue. Considering the variability in initial lesion size, a second recovery metric that removed the effect of lesion size was used. Lesion Closure Rate (LCR) measured the change in re-drawn polygon areas as a relative proportion to the area of the original lesion. LCR was reported as a percentage (%) of the original lesion size per day.

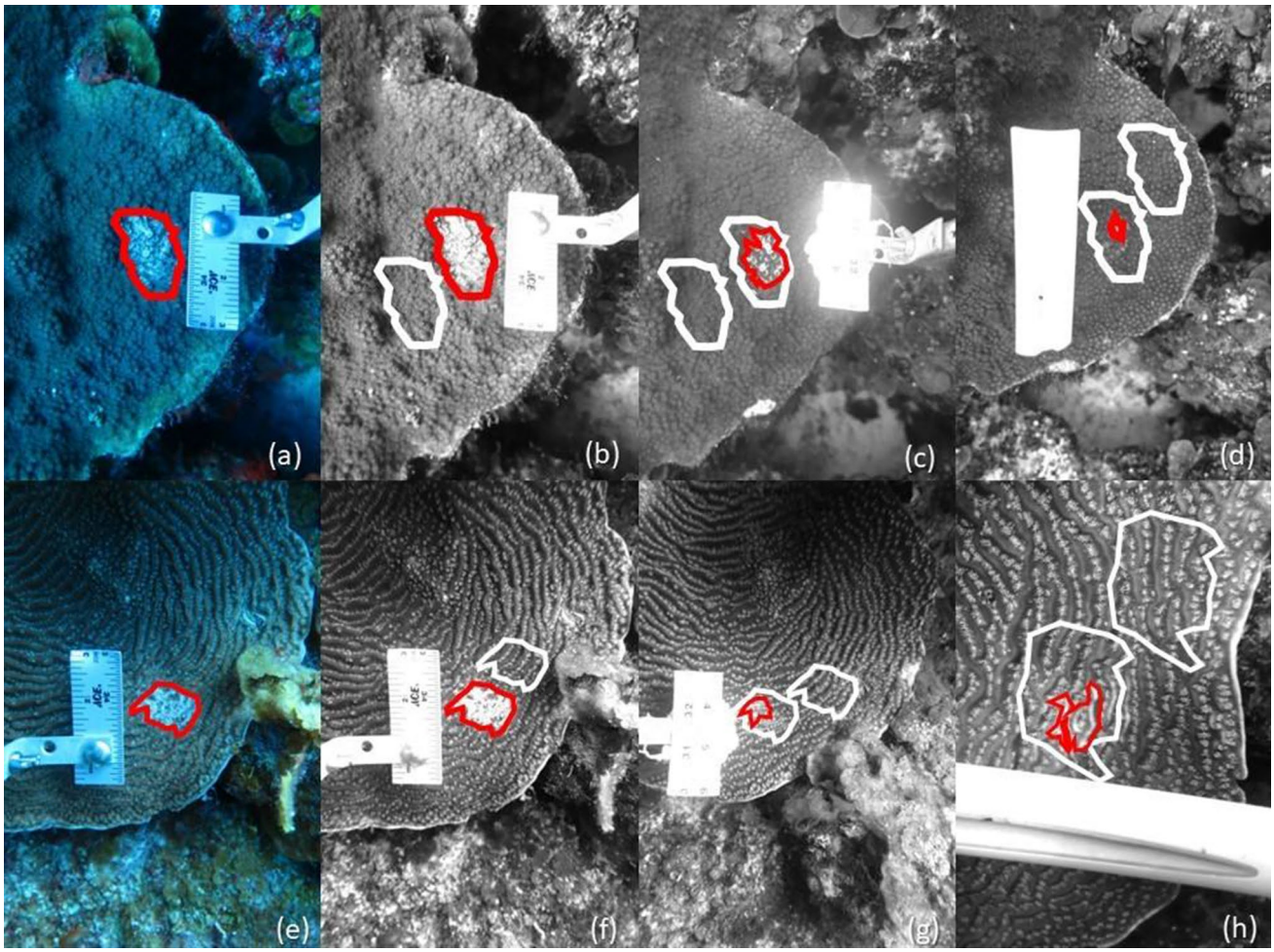

Fig. 1 Sample images of lesion generation and recovery on an *O. franksi* individual (**a**–**d**), and an *A. lamarcki* individual **e**–**h** through time. The first image in each series (a, e) contains an image of the lesion generated by the researchers with a scale bar. The second image in each series (b, f) is the same image that has been scaled to black and white according to the parameters of healthy tissue and lesioned tissue. Subsequent images (c–d, g–h) depict the same coral individuals as lesions recover through time. Red polygons in the image depict the approximate lesion boundaries, used for determining TRR and LCR. White polygons indicate the original lesion area across the time series, used to measure average grayscale value of the lesioned area through time. Second white polygons of the same size and shape are added to measure average grayscale value of healthy tissue of the same area, which is then compared to the lesion area for relative difference in lesion pigment level compared to healthy tissue

Finally, in addition to changes in lesion area, a novel method of lesion observation used pigmentation as a metric for return of healthy tissue to the lesioned area. Following lesion generation and tracing of the lesion using the polygon selection tool, the grayscale of the image was standardized with the eyedropper tool such that pixels with healthy tissue were fully black (color scale 255) and pixels without tissue were fully white (color scale 0). Once standardized, the average grayscale value of color present within the traced lesion area was measured using the measure tool. At the same time, the lesion trace was duplicated and translocated to an area adjacent to the lesion with entirely healthy tissue, where an average grayscale value of color present within the drawn area was also measured. The proportional difference between the two grayscale values (the lesioned area and the healthy area of the same size/shape) was considered the proportional difference between the lesion and healthy tissue in terms of tissue pigment. This process was repeated using the same initial lesion trace at each observation stage, and the change in proportion between healthy tissue and the lesioned area considered the rate that healthy tissue pigment returned to the lesioned area. As this value was similarly relative to the unique lesion on each coral, lesion size and shape is accounted for in the recovery metric. This technique in particular accounted for observations of healthy tissue pigment returning to the middle of lesions, or instances where lesioned polyps appeared partially recovered but not fully pigmented for extended periods. Algal colonization

was not observed on any lesions during the study period and therefore did not impact pigmentation sampling. All calculated recovery rates are summarized in Supplemental Tables (S4–S7).

Calculated recovery rates were then compared in individual models against the noted depth for each individual to determine the effect of increased depth on the rate of recovery from lesions. Recovery rates were also modeled with measured and calculated environmental parameters (temperature and benthic orbital velocity regimes), as well as individual colony characteristics (colony size and initial lesion size).

DNA extraction and amplification of *Symbiodiniaceae*

For each lesioned colony, all tissue removed because of the experimental lesion process was then collected and preserved in ethanol within 4 h of collection. From this tissue, the genetic relatedness of the symbionts present was analyzed using a PSB minicircle amplification technique. Zooxanthellae DNA was isolated using DNA Isolation Buffer in a microcentrifuge tube and resuspended with sodium dodecyl sulfate (SDS) and heated to 65 °C for 30–60 min. Samples were then heated to 45 °C in the presence of Proteinase K for 6 h. If needed, the lysates were then chloroform-extracted once, phenol-extracted twice, and precipitated in ethanol at −20 °C. Precipitates were resuspended in 0.3 M sodium acetate, precipitated again with ethanol, and then resuspended in water and stored −20 °C. The chloroplast minicircle plasmid from each zooxanthellae was then amplified using a polymerase chain reaction (PCR) and when needed further cleaned using Exosap. The product was then submitted for sequencing and analyzed using Molecular Evolutionary Genetic Analysis (MEGA). Successful sequences were then identified using the Blastn database.

Statistical analysis

To explore the direct relationships for each recovery metric and increased depth, multiple linear regression models were built for each species. Models were generated for each recovery metric first considering all measured random variables: temperature regime, benthic orbital velocity regime, colony size, and initial lesion size. Saturated models were then compared to "null" models of standard linear regressions of each recovery metric and depth alone using a Wald's Chi-square comparison to assess for significantly increased model fit with additional terms. Nonsignificant terms in each of the saturated models were then removed for new multiple regression models for each recovery metric, considered "fitted" models. The fitted models were additionally compared to both the respective null models and saturated models for significant differences in model fit using a Wald's Chi-square comparison.

A three-way beta-distribution analysis of all recovery metrics (TRR, LCR, PRR) determined that due to differences in variability of each metric, spherical homoscedascity could not be achieved for a parametric assessment. Therefore, a three-factor permutational analysis of variance (PERMANOVA) was used to assess overall differences for all recovery rates across the factors of species, reef depth zone (above 30 m depth as shallow vs. over 30 m depth as mesophotic), and sampling point.

Statistical analysis was completed using R version 3.5.2. Data importation and manipulation in R was completed using the stringr, dplyr, plyr, ncdf4, and naniar packages. Multivariate analyses were completed using the vegan package. Multiple regression models were generated and tested using the stats package. Figures were generated using the ggplot2, grid, gridExtra, and ggrepel packages.

Results

Orbicella franksi lesion recovery

O. franksi recovery models are summarized in Table 1. Modeled TRR decreased significantly as depth increased ($r^2 = 0.408$, $p = 0.019$), with no significant change to model fit with the addition of any variables ($p = 0.196$), nor were any additional variables significantly associated with TRR. Specifically, corals at mesophotic depths either failed to show lesion recovery or saw increased lesion size after lesion generation (negative regeneration slopes), whereas most colonies at shallower depth (78%) showed lesioned tissue regeneration (Fig. 2). LCR and PRR had no significant relationship with increased depth ($r^2 = 0.025$, $p = 0.563$ and $r^2 = 0.241$, $p = 0.105$, respectively), nor was either model fit significantly increased with the addition of any physical or biological parameters. Of the 24 corals sampled, symbionts from 9 individuals (3 from mesophotic sites, 6 from shallow water sites) were successfully sequenced and all identified as *Cladocopium goreaui*.

Agaricia lamarcki lesion recovery

A. lamarcki recovery models are summarized in Table 2. Modeled TRR and LCR were not affected by changes in depth. All colonies regardless of depth showed regeneration of tissue and lesion closure (Fig. 2). Multiple regression with additional variables significantly increased model fit for TRR (Wald's Chi-square comparison $p = 0.013$), specifically with the addition of initial lesion size ($p = 0.001$), and coral colony size ($p = 0.024$) in the saturated model, with no relationship between either recovery metric and depth. The

Table 1 Slope, significant terms and r^2 value of all linear regressions tested for *O. franksi*

Orbicella franksi recovery model summaries						
Recovery metric	Model type	r^2 value	Model *p*-value	Significant terms	Term *p*-value	Slope of terms
Tissue regeneration rate (TRR)	**Null**	**0.408**	**0.019**	**Depth**		**−0.0007**
	Saturated	0.890	0.097	None		
	Fitted	N/A- No additional terms				
Lesion closure rate (LCR)	Null	0.025	0.563	None		
	Saturated	0.785	0.071	None		
	Fitted	N/A- No additional terms				
Pigmentation recovery rate (PRR)	Null	0.241	0.105	None		
	Saturated	0.842	0.307	None		
	Fitted	N/A- No additional terms				

Only the null model for TRR had any significant relationship between depth and recovery. None of the physical or environmental parameters that were assessed significantly impacted *O. franksi* lesion recovery

Significant regressions (model *p*-value < 0.05) indicated in bold

resulting fitted model of TRR with depth, colony size, and lesion size significantly increased model fit compared to the null model (multiple $r^2 = 0.68$, adjusted $p = 0.003$, Wald's chi-square comparison of null and fitted models $p = 0.002$). In the fitted model, only increased initial lesion size significantly increased TRR (slope = 0.00917, $p = 0.002$). LCR had a significantly improved model in the saturated model (Wald's Chi-square comparison $p = 0.002$), specifically with the significantly positive relationship between LCR and coral colony size ($p = 0.002$, slope = 0.000245). However, isolating depth and coral colony size in a fitted model yielded no significant relationships and the model did not significantly improve in fit from the null model, suggesting that any relationship between colony size and LCR may have been a relic of oversaturation of terms in the saturated model.

In contrast, PRR for *A. lamarcki* did have a significant positive relationship with depth ($r^2 = 0.38$, $p = 0.006$). Compared to *O. franksi, A. lamarcki* recovery rates were only near zero or negative for certain individuals in shallow water sites, rather than at mesophotic depths. Corals from mesophotic locations had some of the highest observed PRR values, suggesting that increased depth ultimately benefitted pigmentation recovery. There was no significant change in model fit with the addition of any other variables. Of the 24 coral symbiont samples, only four *A. lamarcki* (one from each mesophotic location- Hind Bank, Grammanik Bank, College Shoal, and one from the deepest shallow site- South Capella Shallow) successfully sequenced and all symbionts were identified as *C. goreaui.*

Beta-distribution analyses found no overall differences in any lesion recovery metric variances between species ($p = 0.508$) but indicated greater variability for A. lamarcki in shallow sites compared with sites in mesophotic depths ($p = 0.0405$) and during initial recovery compared to later timepoints ($p = 0.0002$). O. franksi recovery rates overall had increased variability at mesophotic locations ($p = 0.0002$) but no difference in variability through time ($p = 0.1056$) (See Figures S2–S3). PERMANOVA multivariate comparison of lesion recovery variables across species, depth category, and timepoint revealed significant differences between the recovery centroids of the two species ($p = 0.001$, permutations = 999), but not across depth category ($p = 0.837$) or timepoint ($p = 0.115$) overall, with no significant interactive terms.

Discussion

The differences in recovery rates between each species were greater than any intraspecific differences across depth category or stage of recovery. Contrasting recovery trends, observed differences in recovery strategies, and overall variability in both shallow sites and MCE sites display the need for each species to be discussed separately and distinctly to understand their relationship with depth and recovery ability.

For *A. lamarcki* there was no major change in tissue regeneration or lesion closure that was directly associated with changing depth, however there was a significant increase in pigmentation recovery rate with depth. It is important to note that this study range does not include the maximum depths for *A. lamarcki,* which has been observed over 60 m deep in the wider Caribbean (Laverick et al. 2017) and as deep as 80 m in the USVI (Smith et al. 2015). Noted in this study and others, *A. lamarcki* maintains thin, plating morphology across its depth range. In shallower depths, colonies were found in shaded areas and hidden away from direct light, while in deeper environments colonies were unshaded and oriented upwards (Helmuth and Sebens 1993; Smith et al. 2010). Laverick et al. (2017) noted that

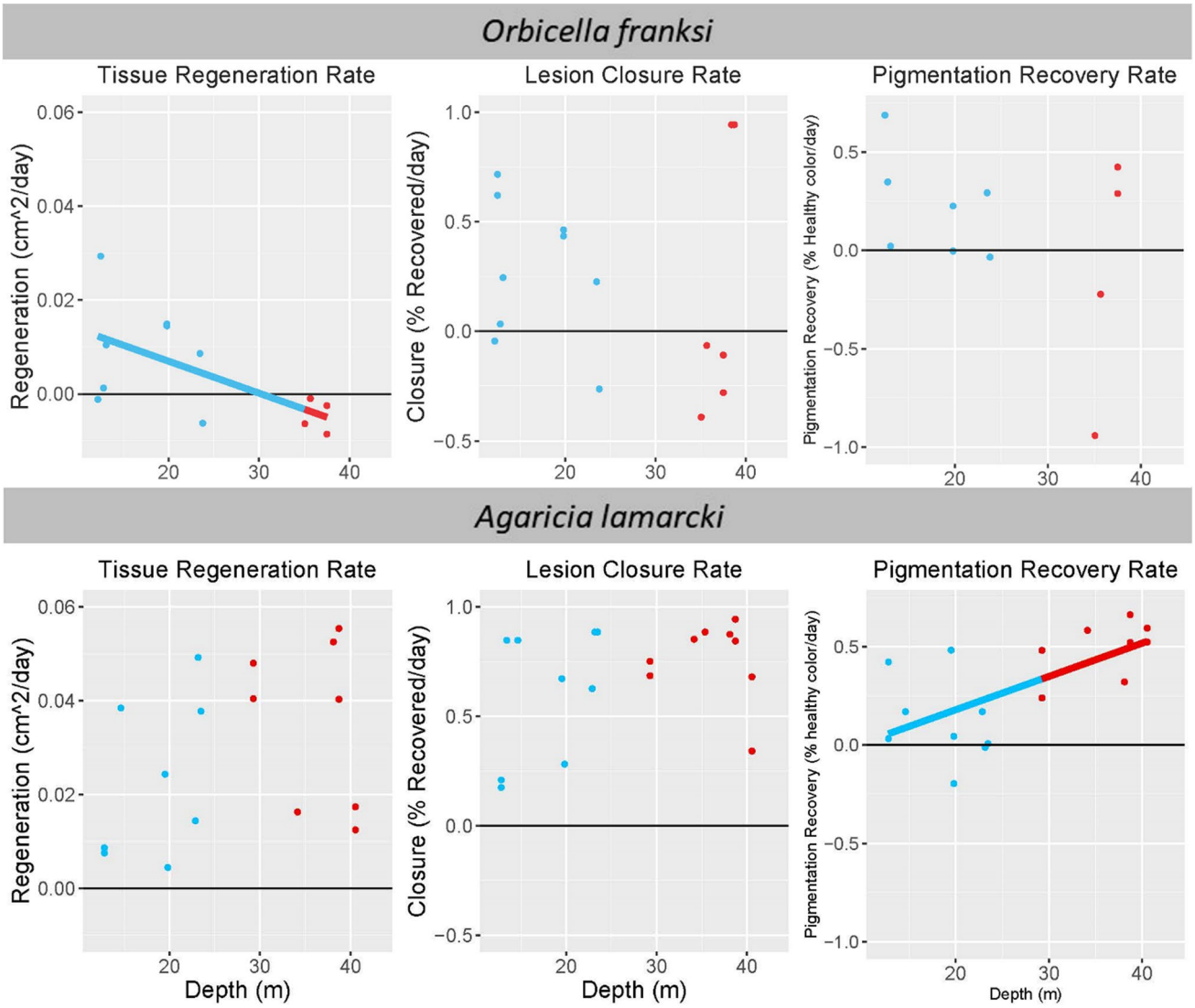

Fig. 2 Tissue regeneration rates, lesion closure rates, and pigmentation recovery rates for *O. franksi* individuals (top) and *A. lamarcki* individuals (bottom). Points indicate actual values, lines indicate best-fit regression modeling outputs. For both points and lines, blue represents individuals in shallow water, red represents individuals in MCEs. For clarity, only significant regression lines are included

while many species may span shallow to mesophotic depths, depth-generalist species like *A. lamarcki* may be less abundant as depth decreases. While these species may survive in shallower environments, they are possibly limited to areas of similar physical conditions as the upper mesophotic such as under ledges due to competitive exclusion (Darling et al. 2012) or excess light or wave energy (Dollar 1982). The physical environment resulting from selective placement on the reef may therefore result in recovery rates and other physiological properties that are comparable to rates seen with deeper conspecifics. The overall higher variability of recovery rates of *A. lamarcki* in shallow water reef sites may be due to these minute differences in the surrounding environment of the colony itself, similar to the observations of Laverick et al. (2019). Differences in physical properties at the colony scale were not assessed in this study but would be an important follow up to better assess the direct relationship of coral recovery to the physical environment.

The depth range of this study did encompass the near-complete habitat range of *O. franksi,* a primary reef-building species less likely to occupy cryptic microhabitats (Carpenter et al. 2008; Smith et al. 2015). The significant decline in TRR as depth increased suggests that some characteristic of depth itself, such as light reduction, reduces this recovery rate for this species. Previous studies of *O. franksi* and other *Orbicella* species found declines in physiological function at the deeper end of this depth range, such as reductions to skeletal growth, calcification, and seasonal energy reserves

Table 2 Slope, significant terms and r^2 value of all linear regressions tested for *A. lamarcki*

Agaricia lamarcki recovery model summaries						
Recovery metric	Model type	r^2 value	Model *p*-value	Significant terms	Term *p*-value	Slope of terms
Tissue regeneration rate(TRR)	Null	0.122	0.186	None		
	Saturated	**0.980**	**0.001**	**Initial lesion Size**	**0.001**	**0.007646**
				Coral colony Size	**0.024**	**0.000009**
	Fitted	**0.683**	**0.003**	**Initial lesion Size**	**0.002**	**0.009170**
Lesion closure rate (LCR)	Null	0.138	0.086	None		
	Saturated	**0.979**	**0.002**	**Coral colony Size**	**0.002**	**0.000245**
	Fitted	0.210	0.216	None		
Pigmentation recovery rate (PRR)	**Null**	**0.424**	**0.006**	**Depth**		**0.015**
	Saturated	0.663	0.544	None		
	Fitted	N/A- No additional terms				

Significant regressions (model *p*-value < 0.05) indicated in bold

(Brandtneris et al. 2016; Groves et al. 2018; Weinstein et al. 2016). In addition, *O. franksi* may undergo mass mortality from intercostal mortality syndrome at its deepest observed depths, a contraction of stressed tissue likely resulting from energy production and acquisition that is lower than metabolic demands (Smith et al. 2010). Therefore, the *O. franksi* depth limit may be driven by lack of available carbon from photosymbionts, which also leads to a notable shift in *O. franksi* morphology at depth to flat, light oriented plates as the depth limit is reached (Cairns 1982; Dustan 1975; Lajeunesse et al. 2018; Todd 2008). In shallow water, *O. franksi* is found as a mounding colony in depths as shallow as 5 m (Carpenter et al. 2008; Smith et al. 2015), suggesting that this species has physiological capacity for environmental conditions associated with shallow reefs, such as higher light, temperature, and wave energy (Lesser et al. 2009).

Observable differences in recovery strategy may also help explain some of the increased variability in all three recovery rates for *O. franksi* individuals sampled in mesophotic depths. On some *O. franksi* individuals at deeper sites, coral polyps recolonized but remained only partially pigmented for an extended period, a common visual defining trait of *O. franksi*. As a result, rates of lesion closure and rates of healthy pigment recovery varied greatly from individual to individual within MCE sites, as some lesions closed quickly with reduced pigmentation while others closed slowly with pigmentation filling in lesion. Both the highest and lowest LCR and PRR rates observed were at the deepest sites, where TRR was lowest. This consistent decline of TRR with increased depth suggests that while strategies in recovery may vary, the overall energetic input available to *O. franksi* individuals (and tissue ultimately created) continuously declines as depth increases.

Temperature and benthic orbital velocity regimes exerted no significant influence on any recovery metric for either species. As no anomalies (temperature or wave energy) were recorded during the study time period, these environmental parameters therefore do not directly relate to recovery physiology and ability outside of stress events. Of the measured biological factors, only larger lesion size increased *A. lamarcki* TRR. This effect was not documented in *O. franksi*, likely due to the overall higher variability in recovery rates. In considering recovery as a relative proportion to lesion size (LCR and PRR), *A. lamarcki* recovery rates were constant across the different sized lesions, only increasing PRR with depth alone. This indicates that variability in lesion size (and importantly, perimeter) may affect the area of tissue that can be regenerated at a given time. The effect of lesion size on recovery has been previously documented in other studies of lesion recovery (Meesters et al. 1996; Oren et al. 1997), but never at the scale of larger lesioned wounds, which require more time to fully heal.

Other factors not assessed in this study likely exerted a strong influence on coral recovery, as indicated by the high variability in lesion recovery rates. These could include differences in genetics, heterotrophy, or even symbiont diversity, which were not fully assessed in this study. Differences in life history traits for each species, such as timing with spawning seasons, may have affected overall energy allocation and possibly reduced recovery rates in each species. Based on predicted spawning times in the Caribbean and USVI, it is likely that this study was conducted during or immediately prior to *O. franksi* spawning in the late summer (Holstein et al. 2015; Szmant 1986) and during or following estimated spring spawning of *A. lamarcki* (Richmond and Hunter 1990; van Moorsel 1983). Brandtneris et al. (2016) noted reduced energy reserves for *A. lamarcki* in July and *O. faveolata* in September (both during this study period) specifically in MCE populations, suggesting that energy availability for recovery could have been reduced for both species in this time specifically for individuals at mesophotic depths.

Variability of lesion recovery rates between *O. franksi* and *A. lamarcki* observed between the two depth zones suggests species-specific depth physiology, such as altered skeletal morphology and energy availability (Enríquez et al. 2005; Terán et al. 2010), plays a key role in differing recovery ability in shallow and mesophotic zones. This study identified direct relationships of depth to recovery rate for both species, and use of a traditional 30 m-depth habitat distinction did not significantly match observed recovery trends for either species. Collectively, this reinforces that characterizing mesophotic habitats using distinct depth limits is overall a poor descriptor for understanding coral physiology, as compared to applying the effect of depth as a gradient. This could be further corroborated by other studies' more recent use of shifted environmental parameters or even ecological shifts, to better define the start of the mesophotic (Laverick et al. 2019; Lesser et al. 2009).

Observed effects of depth on tissue regeneration rate for *O. franksi* indicate that individuals in MCEs may be less resilient to stress events. Historically these deep reefs have been shown to be more sensitive to certain stress events, such as exhibiting higher disease prevalence (Chaves-Fonnegra et al. 2021), and lowered thermal thresholds and higher sensitivity to thermal stress (Smith et al. 2016b, 2019a, b). Despite having high coral cover, MCEs dominated by *O. franksi* may be slower to recover from acute stress events, a particular concern as stony coral tissue loss disease continues to affect shallow and mesophotic reefs in the USVI and across the Caribbean region (Brandt et al. 2021). Overall, this may increase the risk of continuous decline from repeated or ongoing events. By contrast, upper MCEs may serve as an important depth refugia for more resilient *A. lamarcki* populations, shielding the species from some shallow water stressors such as excess light, anomalous wave events, and ecological competition, with consistent or increased recovery following stress events. Considered together, while both species are prevalent throughout the depth range of this study, *O. franksi* populations at depth may be less resilient than *A. lamarcki* populations in the face of increasing frequency and severity of stress events. The continual need to recover from stress events, regardless of depth, will be a key consideration in shaping the reefs of the future.

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Author contributions All authors contributed to the study conception and design. Material preparation, field data collection were completed by JET and TBS. Computational data collection was completed by SM. Data analysis was performed by JET and MB. The first draft of the manuscript was written by JET and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability All DNA sequence data collected during this study is available in Genbank. Remaining datasets generated and analyzed during the current study are included in this publication [and its supplementary information files.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose. On behalf of all authors, the corresponding author states that there is no conflict of interest.

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